Requirements and service boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Requirements and service boundaries before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Neural network experiment defense

Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Requirements and service boundaries trace and record the first step you can explain without help.

Use W2: Trace the worked case for Requirements and service boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Requirements and service boundaries explanation against the supplied

> facts before discussing it.

Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Requirements and service boundaries facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Requirements and service boundaries facts and keep your reasoning.

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Requirements and service boundaries facts and keep your reasoning.

A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Requirements and service boundaries facts and keep your reasoning.

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Requirements and service boundaries facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Requirements and service boundaries to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled requirements and service boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of requirements and service boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Requirements and service boundaries

> Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket. Explain the labelled step before checking the changed condition.

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

Approved card P says 1912. The contract permits reading cards, not selling tickets.

AI ACADEMY / AI-6/1

Requirements and service boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read approved card

The supplied year is 1912.



2. Split the request

Answering the year and selling a ticket are separate operations.



3. Apply the boundary

Return 1912 with card P; do not sell a ticket.

CHANGE ONE THING

Replace P with Q, which has no approved card.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return 1912 with card P; do not sell a ticket.

1. Read approved card

2. Split the request

3. Apply the boundary

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply the boundary Return 1912 with card P; do not sell a ticket. A **service boundary** states permitted work; useful information does not grant extra operations.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace P with Q, which has no approved card. Which conclusion is supported by the supplied facts?

1. The year is unsupported, and the ticket sale is still excluded.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

Repair a missing building block

Review Neural network experiment defense (ai-5/36). Goal: Defend a network experiment using traceable **evidence**. Short practice: Calculate both weighted terms, z and ReLU **output**.

Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

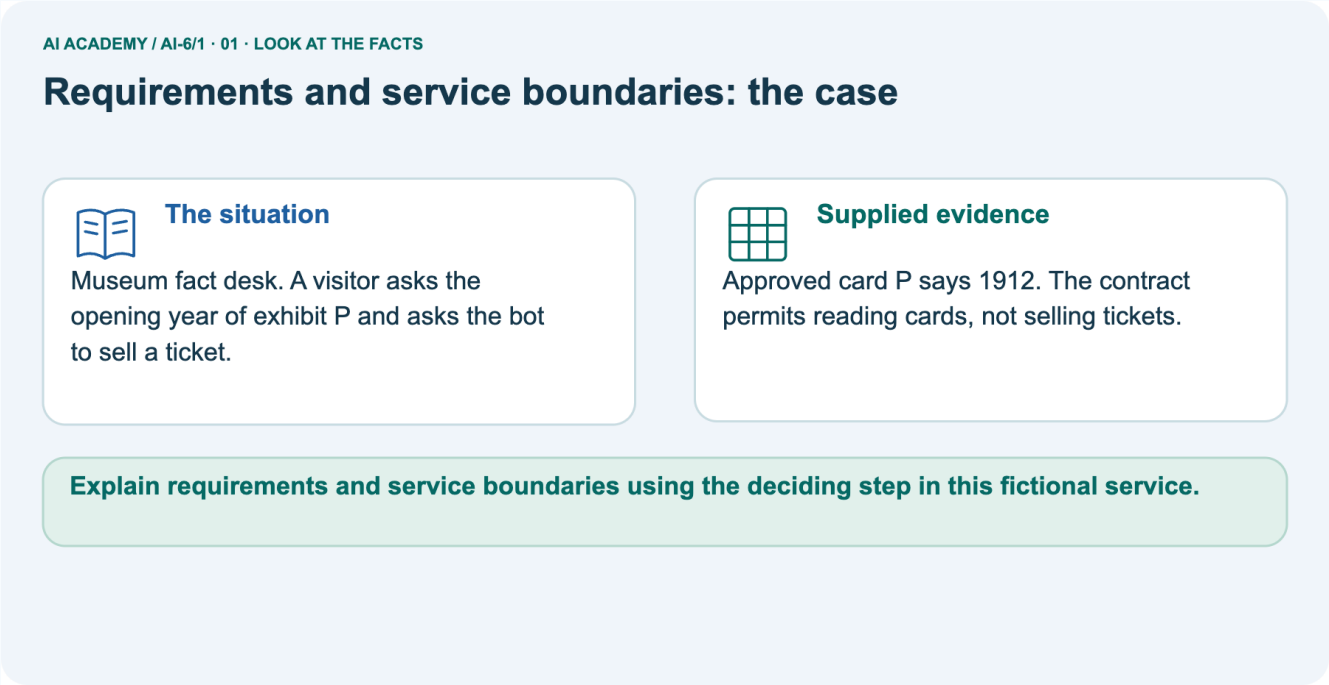
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Requirements and service boundaries picture from its labels, then record a relevant limit.

01. Requirements and service boundaries: the case



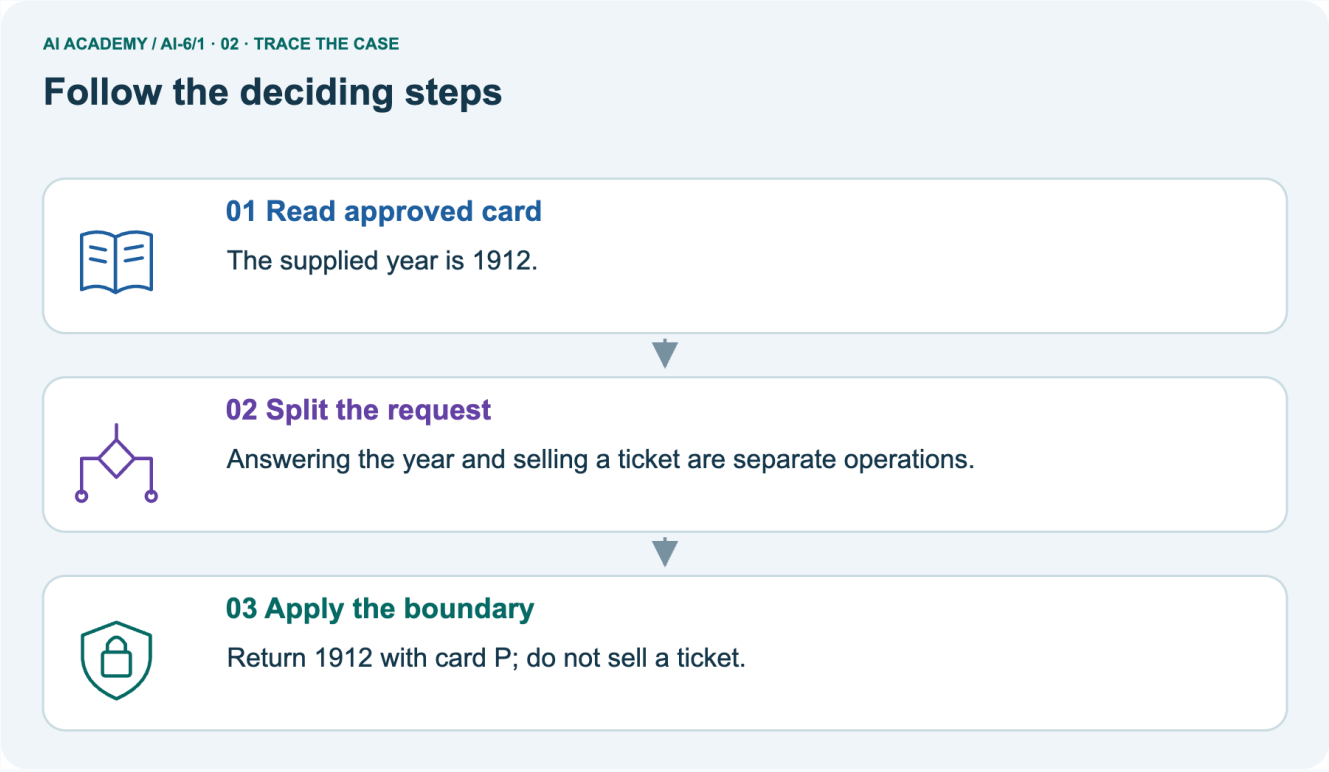
Follow the labelled picture with the matching explanation. Open the vector diagram

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.
Supplied facts: Approved card P says 1912. The contract permits reading cards, not selling tickets.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



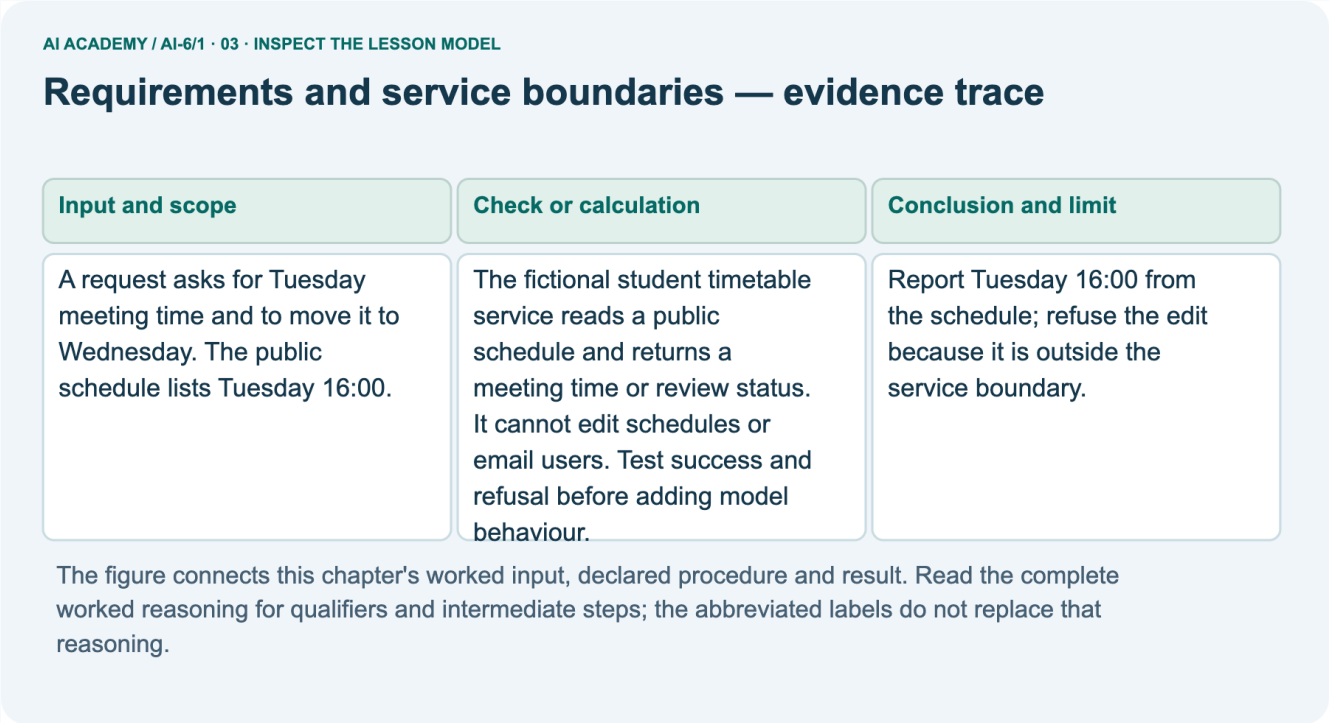
Follow the labelled picture with the matching explanation. Open the vector diagram

Read approved card: The supplied year is 1912. → Split the request: Answering the year and selling a ticket are separate operations. → Apply the boundary: Return 1912 with card P; do not sell a ticket.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Requirements and service boundaries — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements and service boundaries — **evidence** trace. A request asks for Tuesday meeting time and to move it to Wednesday.

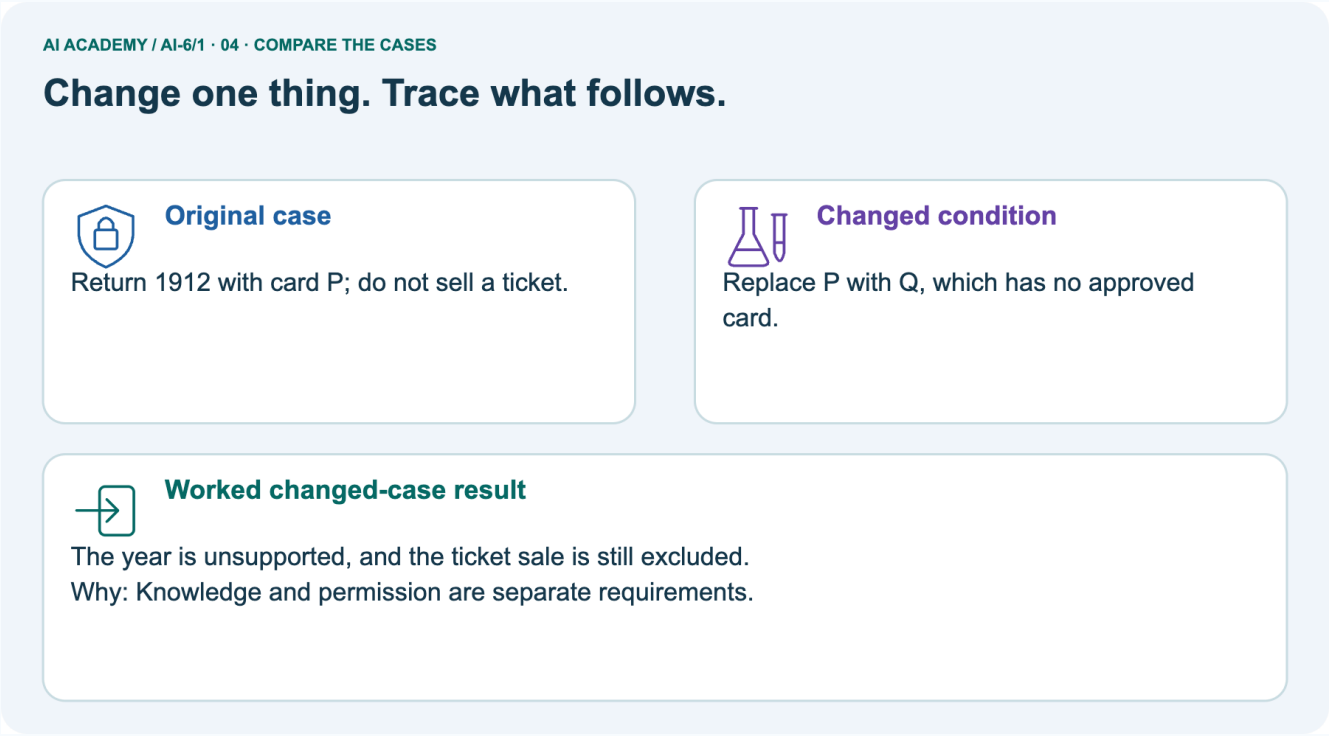
The public schedule lists Tuesday 16:00. | The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users.

Test success and refusal before adding model behaviour. | Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

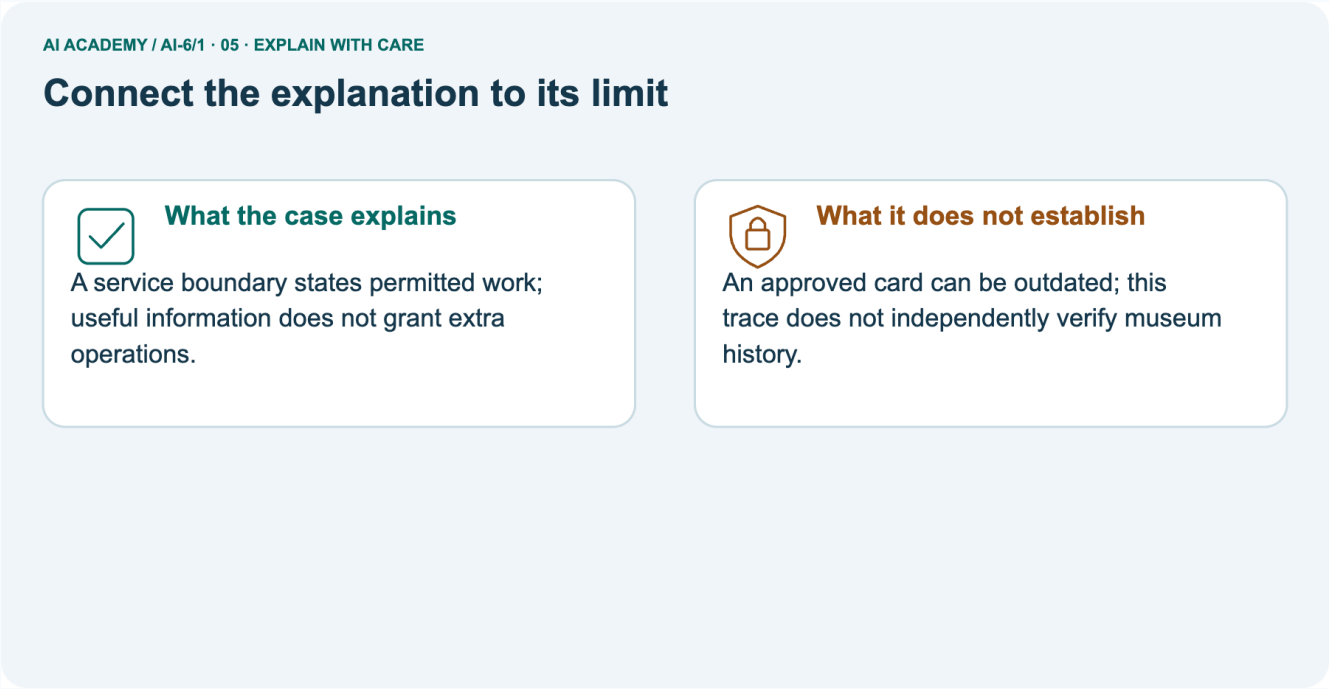
Original case: Return 1912 with card P; do not sell a ticket. Changed condition: Replace P with Q, which has no approved card. Expected result: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



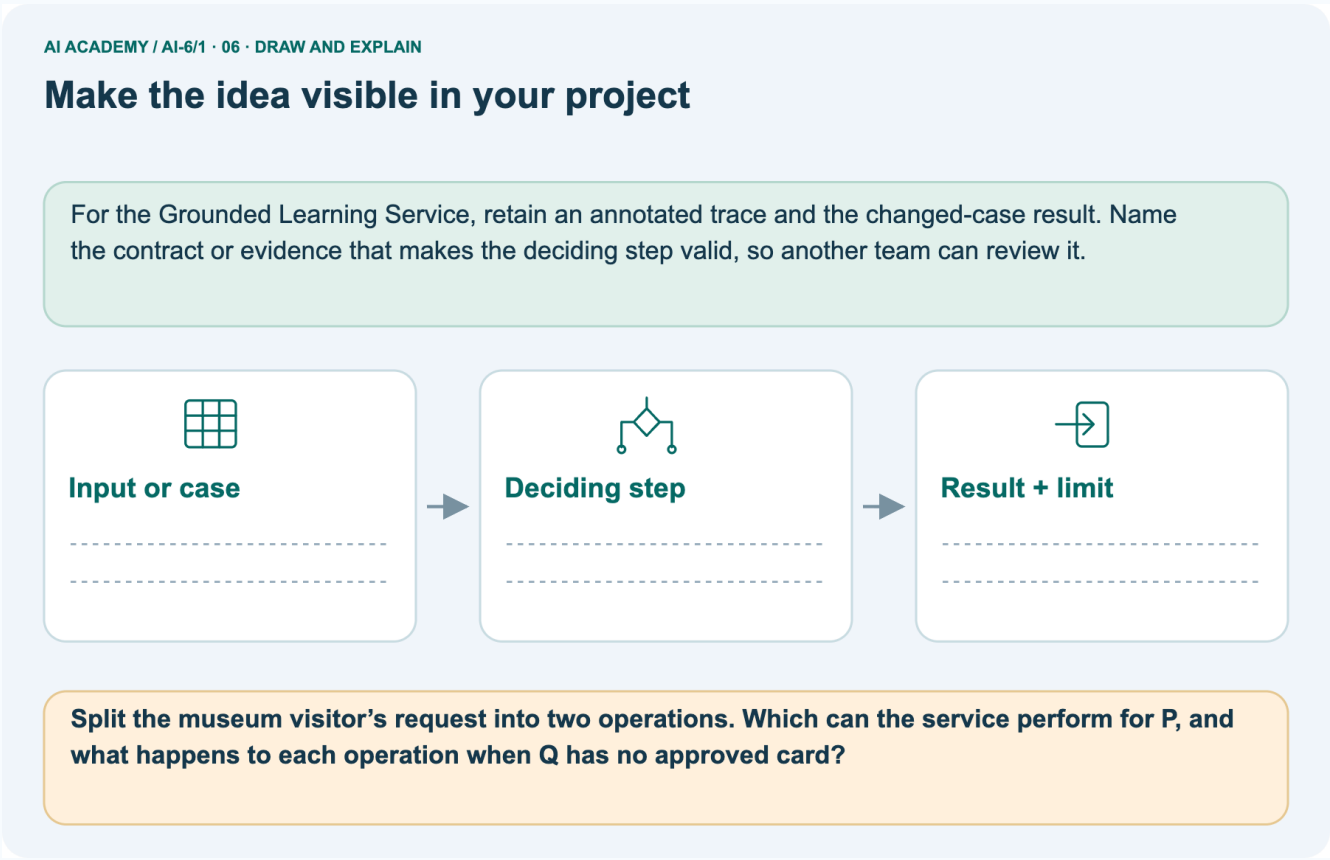
Follow the labelled picture with the matching explanation. Open the vector diagram

A **service boundary** states permitted work; useful information does not grant extra operations. Limit: An approved card can be outdated; this trace does not independently verify museum history.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.